

# Dose Calibrator

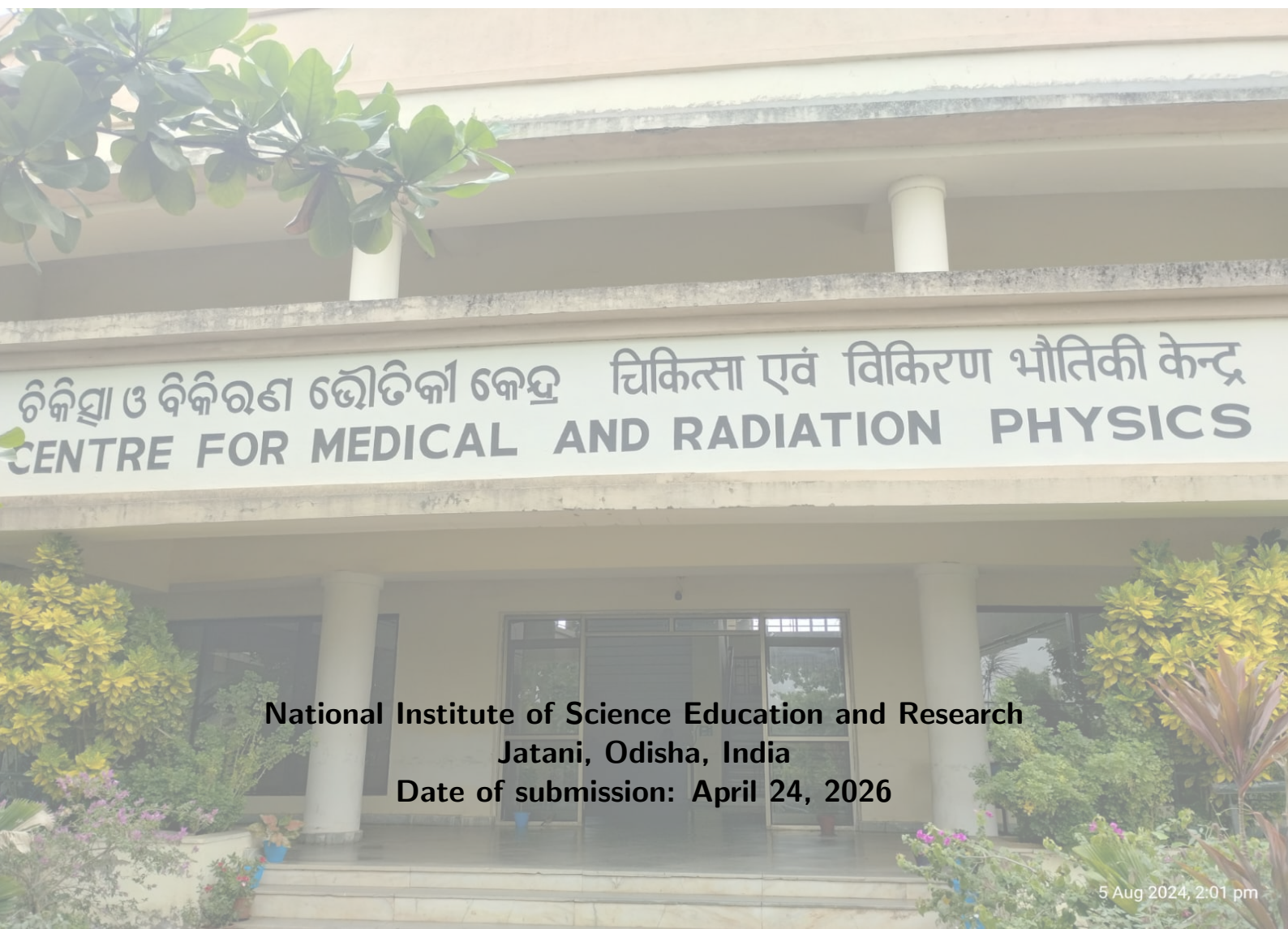


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## 1 Objective

To perform a Quality Assurance (QA) test of the dose calibrator.

## 2 Apparatus

Equipment required:

- Dose calibrator
- Check source
- Canister
- Source holder

## 3 Theory

The dose calibrator is one of the most essential instruments in nuclear medicine for measuring and verifying the radioactivity of radiopharmaceuticals before administration to patients. A typical radioisotope calibrator contains an ionization chamber, a high-voltage power supply, an electronic amplifier, and a display unit to select the radioisotope to be calibrated.

The ionization chamber used for the dose calibrator is generally cylindrical well type in shape and contains Argon gas under high pressure ( $\sim 20\text{--}30$  atm). The geometric goal of the well-type design is to approximate ideal  $4\pi$  geometry. The highly pressurized gas creates favorable conditions for ionization, resulting in high detector efficiency. The calibrator has two coaxial cylindrical electrodes maintained at a voltage difference from a suitable supply.

In the associated electrometer, the ionization current is processed and displayed in units of activity, commonly in digital form. Lead shielding around the ionization chamber protects personnel from radiation hazards and reduces the background contribution in measurement. A sample holder and a removable liner (which can be cleaned in case of contamination) are usually provided.

When a radioactive vial or syringe is placed into the dose calibrator well, the Argon gas gets ionized. Ion pairs migrate toward the anode and cathode, generating a current proportional to radioisotope activity. Since this current is very small (typically in  $\mu\text{A}$  range), a highly sensitive electrometer is required. Output is displayed in mCi or MBq.

### 3.1 Tests Performed

#### 3.1.1 Accuracy Test

The accuracy test signifies how closely an instrument reading matches a known value. A standard source with known activity at a reference date and time is used. The decay-corrected activity at measurement time is calculated and compared with measured activity. Measured values should be within  $\pm 5\%$  of the standard value.

#### 3.1.2 Constancy Test

Constancy is a routine quality control test to ensure reproducibility. A long-lived standard source (for example  $^{137}\text{Cs}$ ) is measured repeatedly (about 5–10 times). Deviation between repeated measurements should be minimal, and values should remain within tolerance.

#### 3.1.3 Linearity Test

This test checks whether measured activity follows expected decay over a wide range (from high activity down to low activity). A source such as  $^{99m}\text{Tc}$  or  $^{18}\text{F}$ -FDG is measured at fixed time intervals, and measured values are compared with expected decay-corrected values.

### 3.1.4 Geometry Test

This test evaluates whether measured activity changes with sample geometry. A known activity is measured while changing source volume (by serial dilution with saline). Ideally, readings should remain nearly constant if the response is geometry independent for the chosen container and setup.

## 4 Observation

### 4.1 Tabulation for accuracy test (Source: $^{137}\text{Cs}$ gel source):

| Initial Activity       | Decay Corrected Activity   | BKG                | Measured Activity    | Net Activity         | % Deviation  | Tolerance |
|------------------------|----------------------------|--------------------|----------------------|----------------------|--------------|-----------|
| 0.211 mCi (18/02/2018) | 174.1084591 $\mu\text{Ci}$ | 1.8 $\mu\text{Ci}$ | 172.5 $\mu\text{Ci}$ | 170.7 $\mu\text{Ci}$ | 1.957664252% | 5%        |

### 4.2 Tabulation for constancy test (Source: $^{137}\text{Cs}$ ):

Initial activity measured = 172.5  $\mu\text{Ci}$

| Repetition No. | Activity Measured ( $\mu\text{Ci}$ ) | Deviation (%) | Tolerance |
|----------------|--------------------------------------|---------------|-----------|
| 1              | 172.5                                | 0.000         | 2%        |
| 2              | 172.2                                | 0.174         | 2%        |
| 3              | 172.5                                | 0.000         | 2%        |
| 4              | 172.4                                | 0.058         | 2%        |
| 5              | 171.9                                | 0.349         | 2%        |
| 6              | 172.6                                | 0.058         | 2%        |
| 7              | 171.6                                | 0.524         | 2%        |
| 8              | 172.1                                | 0.232         | 2%        |
| 9              | 172.4                                | 0.058         | 2%        |
| 10             | 172.5                                | 0.000         | 2%        |

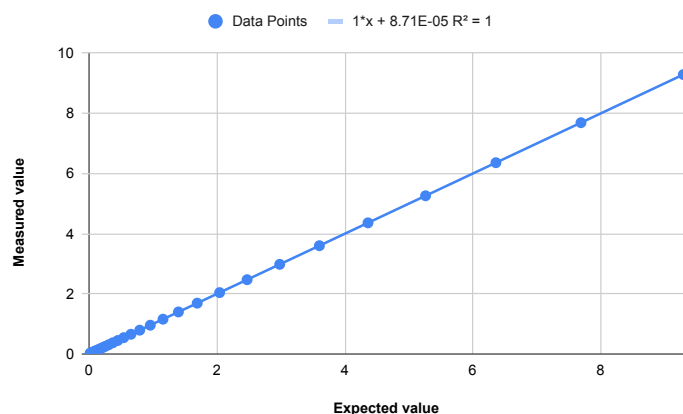
### 4.3 Tabulation for linearity test (Source: $^{18}\text{F}$ -FDG, interval = 30 min):

Table 1: Auto Linearity Test (F-18, Interval = 30 min)

| Date & Time       | Elapsed (min) | Measured  | Predicted | % Var |
|-------------------|---------------|-----------|-----------|-------|
| Apr 20 2026 15:42 | 0             | 9.29 mCi  | 9.29 mCi  | 0.0   |
| Apr 20 2026 16:12 | 30            | 7.69 mCi  | 7.69 mCi  | -0.0  |
| Apr 20 2026 16:42 | 60            | 6.36 mCi  | 6.36 mCi  | 0.0   |
| Apr 20 2026 17:12 | 90            | 5.26 mCi  | 5.26 mCi  | -0.0  |
| Apr 20 2026 17:42 | 120           | 4.36 mCi  | 4.36 mCi  | 0.0   |
| Apr 20 2026 18:12 | 150           | 3.60 mCi  | 3.60 mCi  | -0.1  |
| Apr 20 2026 18:42 | 180           | 2.98 mCi  | 2.98 mCi  | 0.1   |
| Apr 20 2026 19:12 | 210           | 2.47 mCi  | 2.47 mCi  | -0.0  |
| Apr 20 2026 19:42 | 240           | 2.04 mCi  | 2.04 mCi  | 0.1   |
| Apr 20 2026 20:12 | 270           | 1.690 mCi | 1.689 mCi | 0.1   |
| Apr 20 2026 20:42 | 300           | 1.396 mCi | 1.397 mCi | -0.1  |

## Auto Linearity Test (F-18, Interval = 30 min) (Continued)

| Date & Time       | Elapsed (min) | Measured  | Predicted | % Var |
|-------------------|---------------|-----------|-----------|-------|
| Apr 20 2026 21:12 | 330           | 1.155 mCi | 1.156 mCi | -0.1  |
| Apr 20 2026 21:42 | 360           | 956 uCi   | 956 uCi   | -0.1  |
| Apr 20 2026 22:12 | 390           | 791 uCi   | 791 uCi   | -0.0  |
| Apr 20 2026 22:42 | 420           | 654 uCi   | 655 uCi   | -0.1  |
| Apr 20 2026 23:12 | 450           | 541 uCi   | 542 uCi   | -0.1  |
| Apr 20 2026 23:42 | 480           | 448 uCi   | 448 uCi   | -0.1  |
| Apr 21 2026 00:12 | 510           | 370 uCi   | 371 uCi   | -0.1  |
| Apr 21 2026 00:42 | 540           | 307 uCi   | 307 uCi   | -0.1  |
| Apr 21 2026 01:12 | 570           | 253 uCi   | 254 uCi   | -0.2  |
| Apr 21 2026 01:42 | 600           | 210 uCi   | 210 uCi   | 0.1   |
| Apr 21 2026 02:12 | 630           | 173.8 uCi | 173.8 uCi | 0.0   |
| Apr 21 2026 02:42 | 660           | 143.8 uCi | 143.8 uCi | 0.1   |
| Apr 21 2026 03:12 | 690           | 118.8 uCi | 118.9 uCi | -0.1  |
| Apr 21 2026 03:42 | 720           | 98.4 uCi  | 98.4 uCi  | 0.0   |
| Apr 21 2026 04:12 | 750           | 81.5 uCi  | 81.4 uCi  | 0.1   |
| Apr 21 2026 04:42 | 780           | 67.6 uCi  | 67.4 uCi  | 0.4   |
| Apr 21 2026 05:12 | 810           | 56.0 uCi  | 55.7 uCi  | 0.4   |
| Apr 21 2026 05:42 | 840           | 46.5 uCi  | 46.1 uCi  | 0.7   |
| Apr 21 2026 06:12 | 870           | 38.4 uCi  | 38.2 uCi  | 0.6   |
| Apr 21 2026 06:42 | 900           | 31.8 uCi  | 31.6 uCi  | 0.7   |
| Apr 21 2026 07:12 | 930           | 26.4 uCi  | 26.1 uCi  | 1.1   |
| Apr 21 2026 07:42 | 960           | 21.9 uCi  | 21.6 uCi  | 1.2   |
| Apr 21 2026 08:12 | 990           | 18.04 uCi | 17.88 uCi | 0.9   |
| Apr 21 2026 08:42 | 1020          | 15.09 uCi | 14.79 uCi | 2.0   |



**Figure 1:** Linearity test plot for  $^{18}\text{F}$ -FDG source. The measured activity closely follows the predicted decay curve, indicating good linearity of the dose calibrator across a wide range of activities. The percentage variation remains within acceptable limits, confirming the reliability of the instrument for clinical use.

#### 4.4 Tabulation for geometry test (Source: $^{99\text{m}}\text{Tc}$ ):

| Sample Volume | Sample Activity |
|---------------|-----------------|
| 0 ml saline   | 4.41 mCi        |
| 1 ml saline   | 4.38 mCi        |
| 2 ml saline   | 4.39 mCi        |
| 3 ml saline   | 4.41 mCi        |
| 4 ml saline   | 4.43 mCi        |
| 5 ml saline   | 4.42 mCi        |



Mean activity after dilution (from the available observations)  $\approx 4.41$  mCi.

## 4.5 Result

The dose calibrator readings were within tolerance in the performed QA checks:

- Accuracy deviation = 1.957664252% (within 5%).
- Constancy deviations were within tolerance.
- Linearity deviations remained low across the measured range.
- Geometry variation was minimal for the recorded dilutions.

## 4.6 Precautions

- Wear gloves while handling radiopharmaceuticals and the dose calibrator.
- Carefully place source vial/syringe into the calibrator.
- Use L-bench while filling liquid radioisotope/radiopharmaceutical into syringe or vial.
- Wash hands with soap and water after the experiment.
- Always hold source vials using forceps/holding tongs.
- Switch on fume hood while handling evaporative pharmaceuticals.

## 5 Conclusion

The quality assurance tests performed on the dose calibrator indicate acceptable instrument performance for routine nuclear medicine activity measurement. Accuracy, constancy, linearity, and geometry-related checks were found to be within acceptable limits based on observed data.

Hence, regular QA of the dose calibrator is essential to ensure reliable radiopharmaceutical activity measurement and safe patient administration.

### 5.1 References Used in the Experiment Sheet

- Dose Calibrator operation and quality control PDF.
- CRC®-55t (R/PET/W) Radioisotope Dose Calibrator — Owner's Manual.

## References